Antipsychotics

Learning objectives

1. Explain the pharmacological differences between typical and atypical antipsychotic drugs, including their mechanisms of action and receptor affinities.
2. Identify the main clinical indications for the use of typical and atypical antipsychotics in psychiatric disorders.
3. Describe the common side effects associated with both typical and atypical antipsychotics, with special focus on extrapyramidal symptoms (EPS) and metabolic disturbances.
4. Recognize the signs and symptoms of extrapyramidal side effects including acute dystonia, parkinsonism, akathisia, and tardive dyskinesia, and understand their underlying mechanisms.
5. Discuss the nursing role in caring for patients receiving antipsychotic medications, including monitoring for side effects, patient education, medication adherence, and collaboration with the healthcare team.
6. Examine potential drug and food interactions with antipsychotic medications and their clinical implications.
7. Understand contraindications for use of antipsychotic medications and precautions in special populations such as elderly patients with dementia

Introduction

Antipsychotic drugs, also known as neuroleptics, are a class of medications primarily used to manage psychotic disorders such as schizophrenia, bipolar disorder, and severe depression with psychotic features. These medications work by altering the activity of neurotransmitters in the brain, particularly dopamine, serotonin, and, to some extent, norepinephrine. First introduced in the 1950s, typical antipsychotics were the first generation of these drugs. While effective in managing psychotic symptoms, they were associated with significant side effects, particularly extrapyramidal symptoms (EPS) such as tremors, rigidity, and involuntary movements.

The difference between typical and atypical antipsychotics

Item	**Typical**	**Atypical**
Mechanism of action	Block – (antagonize) dopamine receptors in the brain, particularly D2 receptors in the striatum, leading to a decrease in dopamine neurotransmission and a reduction in psychotic symptoms. also block dopamine D1 receptors in the cerebral cortex, which can contribute to their cognitive and motor side effects.	Not only blocks dopamine receptors (D2) but also acts on serotonin 5-HT2A receptors in the *cerebral cortex*, which can contribute to their therapeutic effects on positive and negative symptoms of schizophrenia.
Special	**Clozapine (atypical)** has been shown to have a	

mechanism	higher affinity for dopamine D1 receptors in the *hippocampus*, which may contribute to their effects on cognitive processes. **Risperidone(atypical)** has been shown to have a higher affinity for dopamine D2 receptors in the *amygdala,* which may contribute to their effects on emotional processing	
Binding to receptors	Have a higher binding affinity for dopamine D2 receptors.	Have a lower binding affinity for dopamine D2 receptors.
Dissociate from receptor	Dissociate from dopamine D2 receptors slowly.	Dissociate from dopamine D2 receptors more quickly.
Anticholinergic effect.	Have higher anticholinergic and antihistaminic effects which can lead to higher side effects such as dry mouth and constipation	have fewer anticholinergic and antihistaminic effects compared to typical antipsychotics, which can lead to fewer side effects such as dry mouth and constipation
List of drugs	Chlorpromazine (Thorazine) Haloperidol (Haldol) Fluphenazine (Prolixin) Perphenazine (Trilafon) Thioridazine (Mellaril) Trifluoperazine (Stelazine) Loxapine (Loxitane)	Aripiprazole (Abilify) Clozapine (Clozaril) Olanzapine (Zyprexa) Quetiapine (Seroquel) Risperidone (Risperdal) Paliperidone (Invega) Ziprasidone (Geodon) Lurasidone (Latuda) Asenapine (Saphris)

	Pimozide (Orap)	Iloperidone (Fanapt) Brexpiprazole (Rexulti) Cariprazine (Vraylar)
Duration of use	Not preferred for long-term treatment due to their higher risk of EPS and other motor side effects.	Generally preferred for long-term treatment due to their lower risk of EPS and other motor side effects.
Effect on symptoms	Primarily used to treat positive symptoms of schizophrenia.	Effective against both positive and negative symptoms of schizophrenia.
Indications	**Schizophrenia**: targeting positive symptoms such as hallucinations, delusions, and thought disorders. **Acute Psychosis episodes** in other psychiatric disorders, including bipolar disorder, schizoaffective disorder, and psychotic depression. **Tourette Syndrome**: Some typical antipsychotics, such as pimozide, are used to manage symptoms of	**Schizophrenia**: Atypical antipsychotics are commonly used as first-line treatments for schizophrenia **Bipolar Disorder**: particularly in the management of acute manic or mixed episodes. **Major Depressive Disorder (Adjunctive Therapy)**: Some atypical antipsychotics, such as quetiapine and aripiprazole, are used as adjunctive therapy in

	Tourette syndrome, particularly tics. **Delirium**: In cases of delirium, where there is severe confusion and disorientation, typical antipsychotics may be used to manage agitation and hallucinations. Note (Tourette Syndrome (TS) is a neurological disorder characterized by repetitive, involuntary movements and vocalizations known as tics. These tics typically begin in childhood, often between the ages of 5 and 10, and tend to peak in severity during adolescence before improving in adulthood.	the treatment of major depressive disorder, particularly when antidepressants alone have been ineffective. **Treatment of Resistant Depression**: In cases of treatment-resistant depression, where standard antidepressant medications have failed, certain atypical antipsychotics may be used as augmentation agents to enhance antidepressant effects. **Agitation in Dementia**: Certain atypical antipsychotics, such as risperidone and quetiapine, are used to manage agitation and aggression in elderly patients with dementia, though their use in this population is associated with an increased risk of mortality and other adverse effects.

		Autism Spectrum Disorder (ASD): Some atypical antipsychotics, such as risperidone and aripiprazole, are FDA-approved for the treatment of irritability associated with autism spectrum disorder (ASD) in children and adolescents.
Contraindications	**Parkinson's Disease**: Typical antipsychotics can exacerbate Parkinsonian symptoms, such as tremors and rigidity. **Severe Cardiovascular Disease**: Typical antipsychotics can increase heart rate and blood pressure, which can be problematic for individuals with severe cardiovascular disease. **Severe Liver Disease**: Typical antipsychotics can accumulate in the liver, leading to increased levels of the medication	**Elderly with Dementia**: Atypical antipsychotics are not approved for use in elderly individuals with dementia due to an increased risk of death. **Neutropenia**: Clozapine, an atypical antipsychotic, can cause neutropenia, a loss in the capacity to produce white blood cells needed to fight infection, which can lead to serious infections and potentially death. **Cardiovascular Risks**:

	in the blood, which can be harmful for individuals with severe liver disease. **Severe Kidney Disease**: Typical antipsychotics can accumulate in the kidneys, leading to increased levels of the medication in the blood, which can be harmful for individuals with severe kidney disease.	Atypical antipsychotics can cause changes in heart rate or blood pressure, which can be problematic for individuals with cardiovascular disease. **Metabolic Risks**: Atypical antipsychotics can cause metabolic disturbances, including an increased risk of type 2 diabetes and high cholesterol.
Common Contraindications for Both Typical and Atypical Antipsychotics 1. **Pregnancy and Breastfeeding**: Antipsychotics should be used with caution during pregnancy and breastfeeding due to the potential risks to the fetus or infant. 2. **Allergic Reactions**: Individuals with a history of allergic reactions to antipsychotics should avoid using them. 3. **Severe Allergic Reactions**: Antipsychotics should not be used in individuals with a history of severe allergic reactions, such as anaphylaxis.		

Drug and food interactions with antipsychotics

Antipsychotic drugs can interact with various medications and foods, potentially altering their effectiveness or causing adverse effects. While the specific interactions can vary depending on the type of antipsychotic and individual patient factors, some of the most common drug and food interactions with antipsychotic drugs include:

1. **Central Nervous System Depressants:**

   Combining antipsychotic drugs with other central nervous system depressants, such as benzodiazepines, opioids, or alcohol, can increase the risk of sedation, respiratory depression, and central nervous system depression. This combination should be used with caution and closely monitored.

2. **Anticholinergic Medications:**

   Concomitant use of antipsychotic drugs with other medications possessing anticholinergic properties, such as tricyclic antidepressants, antihistamines, and certain bladder medications, can potentiate anticholinergic side effects such as dry mouth, constipation, urinary retention, and confusion.

3. **Antidepressants:**

   Some antidepressants, particularly selective serotonin reuptake inhibitors (SSRIs) and serotonin-norepinephrine reuptake inhibitors (SNRIs), can increase the risk of serotonin syndrome when used concomitantly with antipsychotic drugs, particularly those with serotonergic activity like risperidone or quetiapine.

4. **Anticonvulsants:**

   Concurrent use of antipsychotic drugs with certain anticonvulsants, such as carbamazepine or phenytoin, can affect the metabolism of

antipsychotics, leading to altered blood levels and potential loss of efficacy or increased risk of side effects. Close monitoring of serum drug levels and clinical response is necessary.

5. **Grapefruit Juice:**

Grapefruit juice contains compounds that can inhibit the activity of cytochrome P450 enzymes in the liver, which are involved in the metabolism of many medications, including some antipsychotics. Consuming grapefruit juice may increase the blood levels of certain antipsychotic drugs, leading to a higher risk of side effects or toxicity.

6. **High-Protein Foods:**

Some antipsychotic drugs, such as aripiprazole, should be taken with or without food, as instructed by the healthcare provider. However, consuming high-protein meals shortly before or after taking these medications may decrease their absorption and effectiveness.

7. **Caffeine**:

Caffeine can potentially exacerbate certain side effects of antipsychotic drugs, such as agitation, anxiety, or insomnia. Patients should be advised to moderate their caffeine intake and monitor for adverse effects.

Patients need to inform their healthcare providers about all medications, supplements, and dietary habits to minimize the risk of drug interactions and ensure the safe and effective use of antipsychotic drugs. Healthcare providers can then make informed decisions regarding medication selection, dosing adjustments, and monitoring to optimize treatment outcomes while minimizing risks.

Common Side Effects of Typical Antipsychotics.

1. **Extrapyramidal Side Effects (EPS):** As acute dystonia, parkinsonism, akathisia, and tardive dyskinesia.

2. **Sedation:** Many typical antipsychotics have sedating properties, which can cause drowsiness, fatigue, and decreased alertness.

3. **Anticholinergic Side Effects**: These include dry mouth, blurred vision, constipation, urinary retention, and cognitive impairment. Anticholinergic effects can exacerbate cognitive deficits in elderly patients and may contribute to falls and confusion.

4. **Orthostatic Hypotension**: Typical antipsychotics can lower blood pressure, leading to dizziness or fainting upon standing up quickly.

5. **Weight Gain**: Some typical antipsychotics, particularly phenothiazines like chlorpromazine and thioridazine, are associated with weight gain, which can increase the risk of obesity and related health problems.

6. **Endocrine Side Effects**: This includes hyperprolactinemia, which can cause menstrual irregularities, galactorrhea (breast milk production), and sexual dysfunction.

7. **Cardiac Effects**: Certain typical antipsychotics, such as thioridazine, can prolong the QT interval, increasing the risk of arrhythmias.

Common Side Effects of Atypical Antipsychotics.

1. **Metabolic Effects:** Atypical antipsychotics are associated with a higher risk of metabolic side effects, including weight gain, dyslipidemia (abnormal lipid levels), and insulin resistance, which can increase the risk of diabetes mellitus.

2**. Sedation:** Some atypical antipsychotics, such as quetiapine and clozapine, can cause sedation and drowsiness.

3. **Orthostatic Hypotension**: Similar to typical antipsychotics, atypical antipsychotics can lower blood pressure, leading to dizziness upon standing.

4. **Anticholinergic Effects:** While less pronounced than with typical antipsychotics, atypical antipsychotics can still cause dry mouth, constipation, and other anticholinergic side effects.

5. **Prolactin Elevation**: Certain atypical antipsychotics, such as risperidone, can cause hyperprolactinemia, leading to menstrual irregularities, galactorrhea, and sexual dysfunction.

6. **QT Prolongation:** Like some typical antipsychotics, certain atypical antipsychotics (e.g., ziprasidone) can prolong the QT interval, increasing the risk of arrhythmias.

7. **Extrapyramidal Symptoms:** Although less common than with typical antipsychotics, some atypical antipsychotics can still cause EPS, particularly at higher doses.

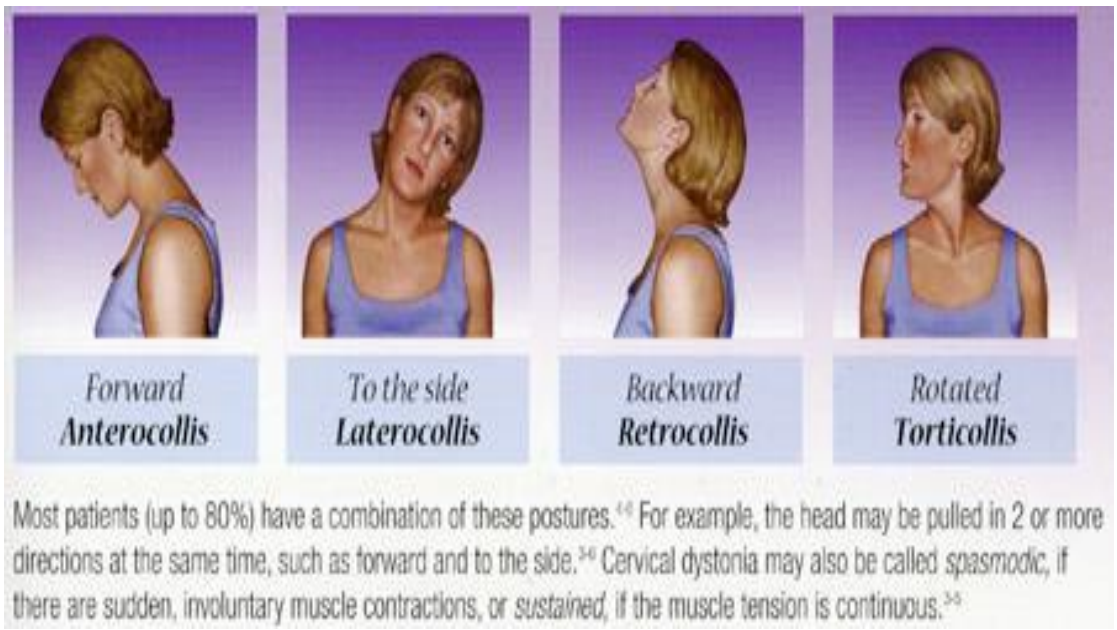
It's important to note that individual responses to antipsychotic medications can vary widely, and not all patients will experience the same side effects. Additionally, the risk of side effects should be weighed against the potential benefits of treatment, and close monitoring by healthcare providers is essential to manage side effects effectively.

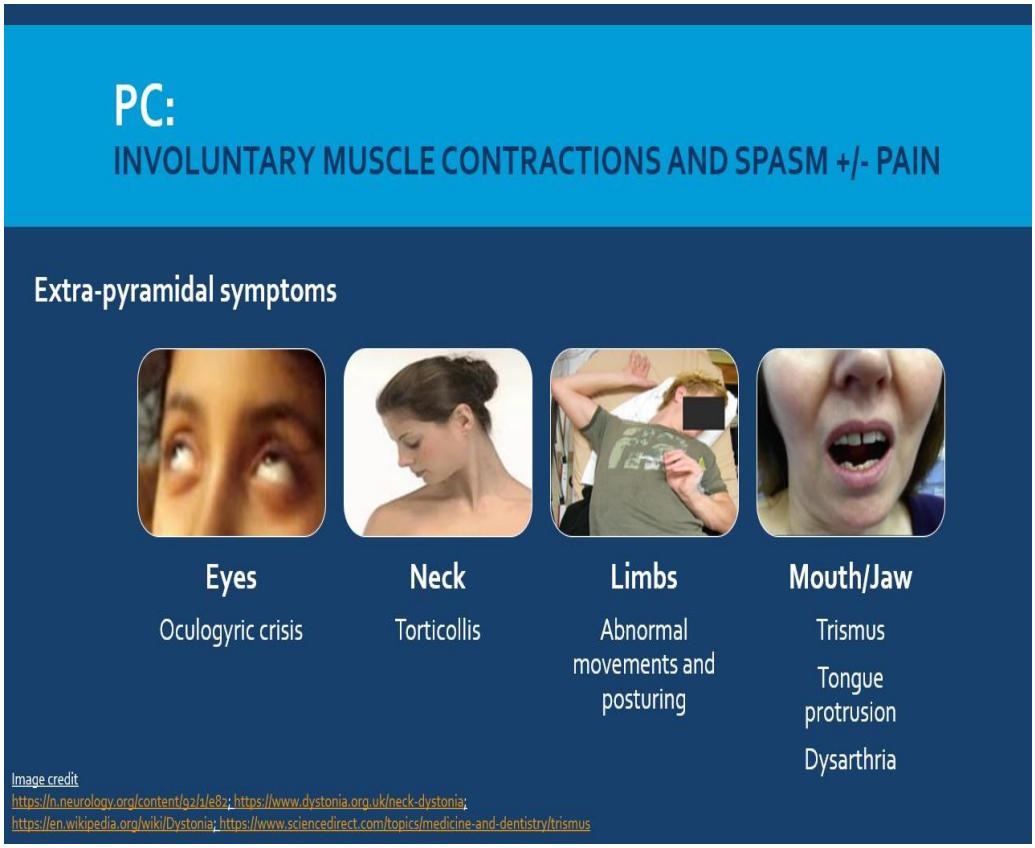
Extrapyramidal side effects

Extrapyramidal side effects (EPS) are a group of movement disorders that are associated with the use of certain medications, particularly antipsychotic drugs. These side effects can range from mild discomfort to severe impairment of motor function and can significantly impact a person's quality of life.

There are several types of extrapyramidal side effects, each with its characteristics and mechanisms:

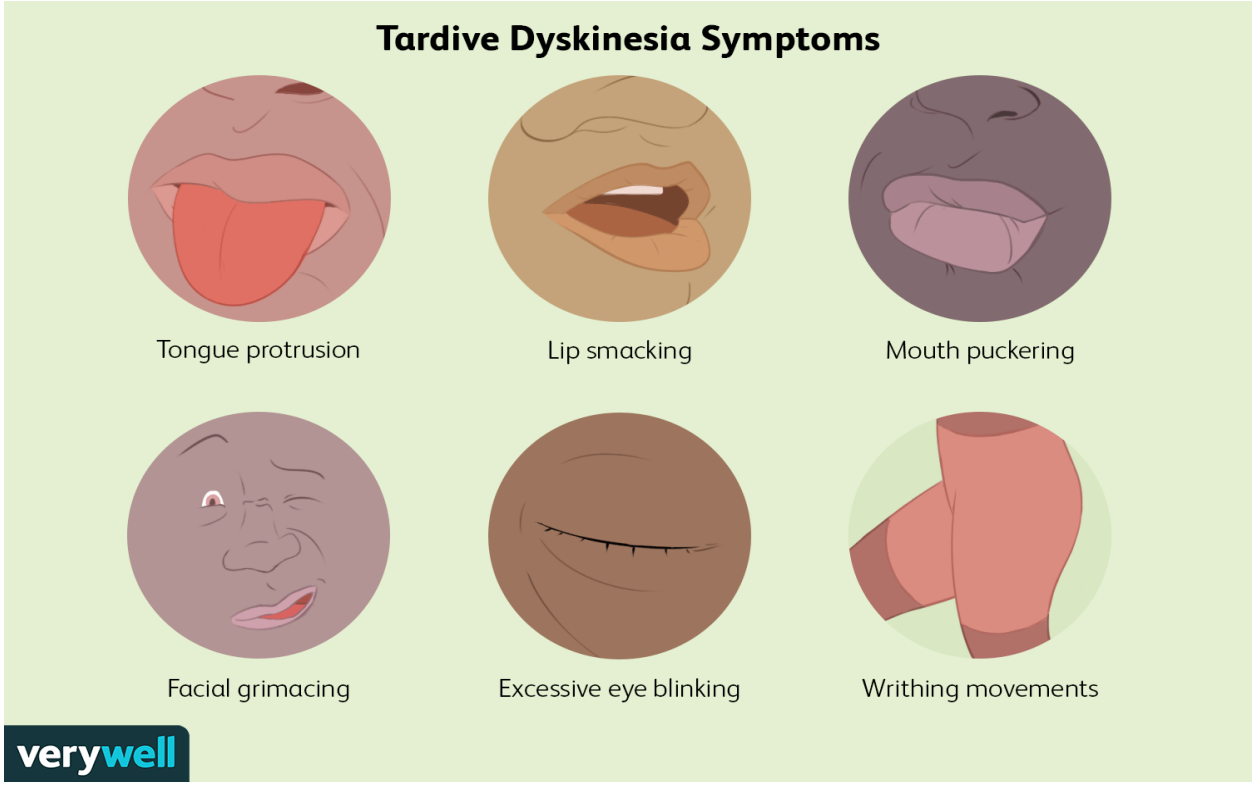
1. **Acute Dystonia:** This is characterized by sudden and sustained muscle contractions, leading to abnormal postures or movements. It often affects the muscles of the face, neck, and tongue, causing symptoms such as grimacing, torticollis (neck twisting), or difficulty speaking or swallowing.

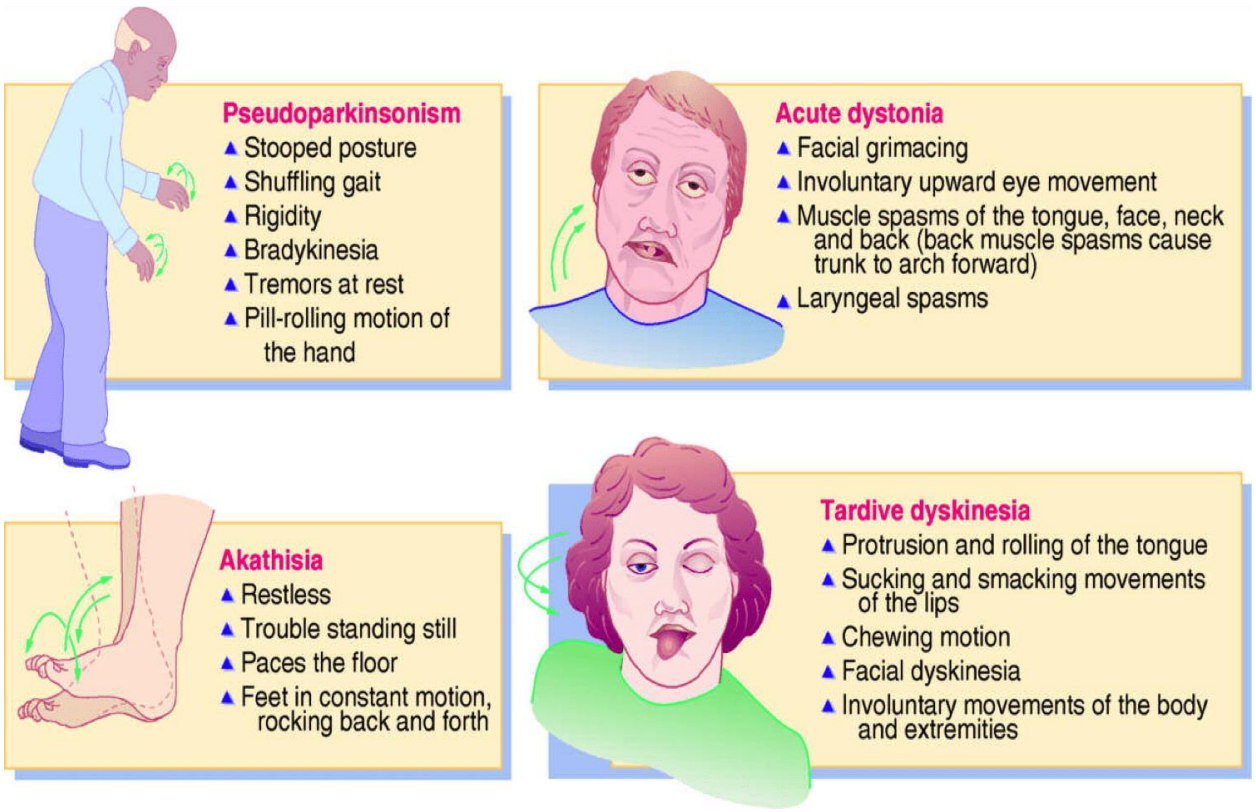




2. **Parkinsonism:** Also known as drug-induced parkinsonism, it resembles Parkinson's disease in its symptoms, including tremors, bradykinesia (slowness of movement), rigidity, and postural instability. These symptoms can significantly impair a person's ability to perform daily activities.
3. **Akathisia:** Akathisia is characterized by an inner sense of restlessness and an inability to sit still. Patients often feel compelled to move their legs or pace around. It can be extremely distressing and is sometimes mistaken for agitation or worsening psychiatric symptoms.
4. Tardive Dyskinesia (TD): This is a potentially irreversible condition characterized by involuntary, repetitive movements, most commonly involving the face and mouth, such as lip smacking, tongue protrusion, or grimacing. TD can be socially

stigmatizing and can persist even after discontinuation of the offending medication.





5. Neuroleptic Malignant Syndrome (NMS): Though rare, NMS is a serious and potentially life-threatening condition characterized by fever, altered mental status, muscle rigidity, autonomic dysfunction (such as changes in blood pressure and heart rate), and elevated levels of creatine phosphokinase (CPK). Prompt recognition and intervention are essential in managing NMS.

The underlying mechanisms of EPS are complex and not fully understood. However, they are believed to involve alterations in the balance of neurotransmitters within the basal ganglia, particularly dopamine and acetylcholine. Antipsychotic medications, particularly first-generation (typical) antipsychotics, block dopamine receptors in the brain, which can lead to dysregulation of motor function and the development of EPS.

Management of EPS typically involves several approaches:

1. **Medication Adjustments**: Switching to a different antipsychotic medication with a lower risk of EPS or adjusting the dose may help alleviate symptoms.

2. **Anticholinergic Medications**: These drugs, such as benztropine or trihexyphenidyl, can help counteract the effects of dopamine blockade and reduce EPS symptoms.

3. **Beta-blockers or Benzodiazepines**: These medications may be used to manage symptoms of akathisia or acute dystonia.

4. **Clozapine:** In cases where other antipsychotics have failed or caused intolerable side effects, clozapine, a second-generation (atypical) antipsychotic, may be considered. It has a lower risk of causing EPS but carries other potential side effects that require monitoring.

5. **Behavioral Interventions**: Psychosocial interventions, such as patient education, supportive therapy, and stress reduction techniques, can help patients cope with EPS symptoms and improve overall treatment adherence.

Prevention of EPS involves careful selection of antipsychotic medications based on the patient's individual risk factors and monitoring for early signs of side effects.

Regular follow-up appointments with healthcare providers are essential for ongoing assessment and management of EPS.

Additionally, educating patients and caregivers about the signs and symptoms of EPS can help facilitate early intervention and prevent potential complications.

Nursing role with patients receiving antipsychotic drugs

Nurses play a pivotal role in the care of patients receiving antipsychotic drugs, particularly those with psychotic disorders such as schizophrenia, bipolar disorder, or severe depression with psychotic features. Their responsibilities encompass various aspects of patient care, including assessment, medication administration, monitoring for side effects, education, and support. Here's a breakdown of the nursing role in patients receiving antipsychotic drugs:

1. **Assessment:**

   Nurses conduct comprehensive assessments of patients before and during antipsychotic therapy.

   This includes gathering information about the patient's medical history, psychiatric symptoms, current medications, allergies, and any past experiences with antipsychotics.

   Nurses also assess the patient's mental status, behavior, and level of functioning to monitor the effectiveness of treatment.

2. **Medication Administration:**

   Nurses are responsible for administering antipsychotic medications according to the prescribed dosage, route, and schedule.

   They ensure that patients receive their medications on time and monitor for any adverse reactions or drug interactions.

   Nurses may also educate patients and their families about the importance of medication adherence and guide how to take the medication properly.

3. **Monitoring for Side Effects:**

Nurses closely monitor patients for potential side effects of antipsychotic drugs, such as extrapyramidal symptoms (EPS), sedation, metabolic changes, and cardiovascular effects.

They conduct regular assessments of vital signs, neurological status, and laboratory values to detect any adverse reactions promptly.

Nurses also observe for signs of tardive dyskinesia (TD), a potentially irreversible movement disorder associated with long-term use of antipsychotics.

4. **Education and Counseling**:

Nurses play a key role in educating patients and their families about antipsychotic medications, including their purpose, dosage, potential side effects, and the importance of treatment adherence.

They provide information on lifestyle modifications, such as diet and exercise, to minimize the risk of metabolic side effects.

Nurses also offer counseling and support to help patients cope with their psychiatric symptoms, address concerns, and promote self-management strategies.

5. **Collaboration with Healthcare Team**:

Nurses collaborate with other members of the healthcare team, including psychiatrists, psychologists, social workers, and pharmacists, to ensure comprehensive care for patients receiving antipsychotic drugs.

They participate in interdisciplinary treatment planning, communicate changes in the patient's condition or medication regimen, and coordinate follow-up care and referrals as needed.

6. **Advocacy and Support:**

Nurses advocate for the needs and preferences of patients receiving antipsychotic drugs, promoting their rights to informed consent, privacy, and dignity.

They provide emotional support, reassurance, and encouragement to patients who may experience stigma, isolation, or challenges related to their mental health condition and treatment.

In summary, nurses play a multifaceted role in the care of patients receiving antipsychotic drugs, encompassing assessment, medication administration, monitoring, education, counseling, collaboration, advocacy, and support. Their holistic approach contributes to the safe and effective management of psychiatric disorders and helps patients achieve improved outcomes and quality of life.

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